

Muhammad Azeem Bhatti

AI/ML Engineer · Applied Research

Islamabad, Pakistan +92 333 7554665 muhammadazeembhatti.ai@gmail.com
[linkedin.com/in/muhammad-azeem-bhatti](https://www.linkedin.com/in/muhammad-azeem-bhatti) github.com/Muhammad-Azeem-Bhatti
orcid.org/0009-0001-9744-4235

EDUCATION

MS, Artificial Intelligence

Air University, E-9 Campus, Islamabad

Sep 2025 – Present · CGPA 3.97/4.0

BS, Electrical Engineering (Electronics)

Air University, E-9 Campus, Islamabad

Sep 2015 – June 2019

EXPERIENCE

AI Intern — GoldenGateInnovations

Jul 2026 – Present

Speech Transcription & Translation Pipeline (Urdu–English)

- Built a production speech transcription/translation pipeline for Urdu-English code-switched audio (faster-whisper large-v3 ASR + Fireworks GLM-5.2 LLM translation), cutting translation WER from ~62% to ~46% via a custom translate-critique-refine pipeline with tuned prompts and retry/batching logic.
- Validated the system against a human-annotated Urdu-English benchmark, achieving 28.9% WER as an externally-verifiable accuracy metric.
- Added real-time speaker diarization (pyannote.audio) to the GPU inference pipeline, resolving production issues spanning cloud model-access restrictions, cross-GPU CUDA incompatibility, and NumPy/PyTorch ABI conflicts.
- Built a bilingual PDF transcript export handling right-to-left text shaping and script detection, and deployed/maintained the GPU inference service on RunPod with Docker and zero-downtime rollouts.

OSINT News Intelligence Pipeline (Pakistan-Interest News Monitoring System)

Python, SQLite, SQLAlchemy, Streamlit, Groq LLM API, RSS/PRAW, APScheduler

- Built an end-to-end OSINT pipeline in Python collecting, deduplicating, and classifying news from 8+ live sources (RSS, Reddit, government portals) into a structured SQLite database.
- Integrated an LLM classifier (Groq API) to tag articles by genre and national-interest alignment, producing structured JSON output with confidence scores across 500+ articles.
- Designed a pluggable source-connector architecture for adding new sources without touching the core pipeline, with content-hash deduplication and APScheduler-driven scheduled batch processing.

PROJECTS

Effectiveness of Balancing Plant's Dataset using GAN

Semester Project

Used a DCGAN to synthetically balance a crop leaf disease image dataset, generating additional samples for underrepresented classes ahead of downstream classification.

Privacy-Preserving Techniques in Federated Learning

Semester Project

Built a PyTorch-based federated learning pipeline combining Hensel's compression, differential privacy on model weights, and secure aggregation to protect client data throughout distributed training.

SKILLS

LANGUAGES

Python

ML/DL & RESEARCH

PyTorch

GANs (DCGAN)

U-Net & Attention

ANFIS / Neuro-Fuzzy

Federated Learning

Differential Privacy

NSGA-II

Harris Hawks Opt.

Explainable AI

Radiomics

Medical Image Segmentation

NLP & SPEECH AI

LLM Integration & Prompting

faster-whisper (ASR)

Speaker Diarization

Machine Translation

TOOLS

PyCharm

VS Code

GitHub

PUBLICATIONS

Multi-Objective Selection of Hybrid Deep–Radiomic Features for Brain Tumor MRI Classification with Transparent Fuzzy Rule-Based Inference

Information Fusion (Elsevier) · First Author · Submitted, with editor

Proposed OHHO-NSGA-II — a three-objective, Opposition-Based Harris Hawks-driven extension of NSGA-II — paired with an ANFIS neuro-symbolic classifier for inherently interpretable brain tumor MRI classification. Reduced a 2,420-feature hybrid pool to 25 (98.97% search-space reduction), reaching 94.45% cross-validation accuracy and 0.9383 macro-F1 on the Figshare MRI dataset.

A Privacy-Preserving Federated Learning Framework with Variance-Based Aggregation and Customized U-Net for Brain Tumor Segmentation

Frontiers in Oncology · Third Author · Ready to submit

Designed Fed-PLM, a federated learning framework pairing a lightweight attention-based U-Net with a novel aggregation method that identifies the dominant latent principal mode across client updates, enabling privacy-preserving, multi-institution brain tumor segmentation on BraTS 2021.